

EE 364B Final Report: Robust Model Predictive Control for Tax-smart Transition

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1 Introduction

Portfolio transition is the problem of reallocating an investor portfolio—a collection of tax lots each carrying a per-share cost basis—toward a target model over multiple periods. The decision is tax-aware: selling appreciated lots triggers capital gains while selling depreciated lots harvests losses, and the optimal trade-off depends on an unknown future price path.

We address this with three modular layers: (i) a multi-period MILP `StoxOptimizer` with a lexicographic objective over terminal tracking, intermediate tracking, and discounted realized tax cost; (ii) a Robust Model Predictive Control `RMPCController` that generates price scenarios via block bootstrap [1] and solves a single two-stage stochastic program [3] at $t = 0$; and (iii) a receding-horizon `Backtester` that executes only the first-stage trade at each step and advances along realized prices. Relative to the midterm report, the model adds a temporal discount factor γ on later-period tax cost, and the experiments sweep γ together with the intermediate tracking band δ to diagnose the closed-loop behavior. The full implementation is available as an open-source package.¹

2 Multi-period MILP Optimization Model

The optimizer is a multi-period MILP organized around a filtration structure that tracks tax lots at the share level, registering a lexicographic objective over terminal tracking, intermediate tracking, and discounted realized tax cost. Specific-identification on a per-share cost basis keeps realized tax cost linear in the sell quantities, and the hierarchy reflects investor priorities: arrive at target, stay within band, then minimize tax subject to both.

Indices and parameters. The core concepts are the filtration and the lot dynamics (Figure 1). Let $f \in \{1, \dots, T\}$ index monthly filtrations, $s \in \{1, \dots, S\}$ index scenarios, k index tickers in the universe $\mathcal{K} = \text{positions} \cup \text{model}$, and (i, j) identify a tax lot whose acquisition period is column j and row index i . Starting lots occupy $j = 0$ with $i \in \{0, \dots, L - 1\}$; purchased lots occupy $j \geq 1$ with $i \in \{0, \dots, N - 1\}$. The lot set available at filtration f is $\mathcal{L}_f = \{(i, 0)\}_{i=0}^{L-1} \cup \bigcup_{j=1}^f \{(i, j)\}_{i=0}^{N-1}$, with $|\mathcal{L}_f| = L + fN$. The subset of *existing* lots—those available to sell at f —is $\mathcal{E}_f = \mathcal{L}_f \setminus \{(i, f)\}_{i=0}^{N-1}$ (all lots except the just-purchased column). Parameters are price $p_{k,f}^{(s)}$, target weight w_k^* , flat capital-gains tax rate τ , intermediate per-ticker band δ , per-share cost basis $c_{i,j}^{(s)}$, and a temporal discount factor $\gamma \in (0, 1]$ applied to later-period tax cost. For starting lots, $c_{i,0}$ is the position’s recorded basis per share; for purchased lots, $c_{i,j}^{(s)} = p_{k(i,j),j}^{(s)}$, the price at which the lot was opened.

¹<https://github.com/trank4/EE-364B-Stochastic-Programming-Tax-Smart-Transition>

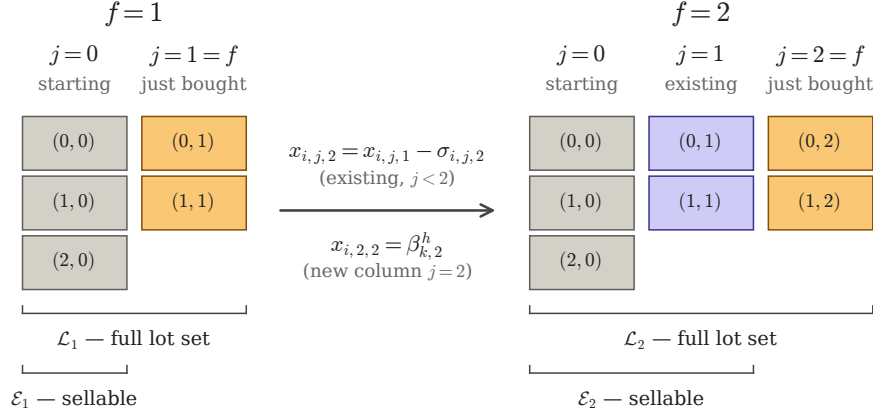


Figure 1: Lot structures and dynamics at filtration $f = 1$ and $f = 2$. A cell (i, j) is a tax lot with acquisition period j and within-column index i . Gray: starting lots; purple: existing lots in $\mathcal{E}_f$; amber: just-bought column $j = f$, excluded from $\mathcal{E}_f$. Each filtration graduates the prior amber column to purple and appends a new amber column. Drawn for $L = 3$, $N = 2$.

Decision variables. All variables are denominated in shares and indexed per scenario. Lot-level: $x_{i,j,f}^{(s)} \in \mathbb{R}_+$ shares held, $\sigma_{i,j,f}^{(s)} \in \mathbb{R}_+$ shares sold. Holding-level: $h_{k,f}^{(s)}, \sigma_{k,f}^h, \beta_{k,f}^h \in \mathbb{R}_+$ for total shares held, sold, and bought; binaries $z_{k,f}^s, z_{k,f}^b \in \{0, 1\}$ indicate whether any sell or buy occurs. Upper bounds for decision variables are derived from a heuristic compounded-AUM cap $\bar{V}_f^{(s)} = V_0 \prod_{t=1}^f \max_k (p_{k,t}^{(s)} / p_{k,t-1}^{(s)})$, assuming the portfolio AUM grows at the highest return among tickers.

Constraints. The scenario superscript is suppressed below; all constraints hold per s .

Lot-holding linking and *sell aggregation* connect lot-level and ticker-level variables:

$$h_{k,f} = \sum_{(i,j) \in \mathcal{L}_f(k)} x_{i,j,f}, \quad \sigma_{k,f}^h = \sum_{(i,j) \in \mathcal{E}_f(k)} \sigma_{i,j,f}. \quad (1)$$

Big-M indicator linking, buy/sell exclusivity, and per-period dollar self-financing:

$$\sigma_{k,f}^h \leq M_{k,f} z_{k,f}^s, \quad \beta_{k,f}^h \leq M_{k,f} z_{k,f}^b, \quad z_{k,f}^s + z_{k,f}^b \leq 1, \quad \sum_k p_{k,f} \sigma_{k,f}^h = \sum_k p_{k,f} \beta_{k,f}^h. \quad (2)$$

The first two link total shares sold and bought to their associated binary indicators. The third blocks within-period simultaneous buy and sell of the same ticker, since the IRS wash-sale rule “washes” away the tax-loss-harvesting benefit in such cases. The fourth enforces self-financing—the portfolio holds no cash, so within each period sell dollars must equal buy dollars (a zero-action period is permitted).

Lot dynamics. Lot shares $x_{i,j,f}$ represent the *post-decision* state at filtration f , after the buy and sell at f have occurred. For existing lots ($j < f$), the post-decision shares evolve from the prior-period shares:

$$x_{i,j,f} = \tilde{x}_{i,j,f} - \sigma_{i,j,f}, \quad \tilde{x}_{i,j,f} = \begin{cases} \text{shr}_i & f = 1, j = 0, \\ x_{i,j,f-1} & f \geq 2. \end{cases} \quad (3)$$

The input share count shr_i serves as the prior-period quantity for starting lots. For newly created lots at column $j = f$, the buy at filtration f initializes the lot directly:

$$x_{i,f,f} = \beta_{k(i,f),f}^h, \quad i = 0, \dots, N-1. \quad (4)$$

Combined with $x_{i,j,f} \geq 0$, the existing-lot identity implicitly enforces $\sigma_{i,j,f} \leq \tilde{x}_{i,j,f}$, i.e. cannot sell what you do not hold.

Objectives. Three objectives are registered lexicographically via Gurobi’s multi-objective interface (`setObjectiveN`), each averaged uniformly over the S scenarios. Higher priority is optimized first; lower priority is minimized subject to the higher priorities staying optimal. Let $V_f^{(s)} = \sum_k p_{k,f}^{(s)} h_{k,f}^{(s)}$ denote the per-scenario portfolio value.

Priority 2 (highest) — Terminal deviation. At the final filtration $f = T$, auxiliaries $\xi_k^{(s)} \geq 0$ linearize the absolute dollar deviation from target weights:

$$\xi_k^{(s)} \geq p_{k,T}^{(s)} h_{k,T}^{(s)} - w_k^* V_T^{(s)}, \quad \xi_k^{(s)} \geq w_k^* V_T^{(s)} - p_{k,T}^{(s)} h_{k,T}^{(s)}. \quad (5)$$

The objective is $J_2 = \frac{1}{S} \sum_{s=1}^S \sum_k \xi_k^{(s)}$.

Priority 1 — Intermediate deviation. At each intermediate post-decision state $f \in \{1, \dots, T-1\}$, only the overshoot beyond the $\pm\delta$ weight band is penalized. Auxiliaries $\zeta_{k,f}^{(s)} \geq 0$ capture the out-of-band dollar deviation:

$$\zeta_{k,f}^{(s)} \geq p_{k,f}^{(s)} h_{k,f}^{(s)} - (w_k^* + \delta) V_f^{(s)}, \quad \zeta_{k,f}^{(s)} \geq (w_k^* - \delta) V_f^{(s)} - p_{k,f}^{(s)} h_{k,f}^{(s)}. \quad (6)$$

Inside the band both right-hand sides are nonpositive, so $\zeta_{k,f}^{(s)} = 0$ is feasible. The objective is $J_1 = \frac{1}{S} \sum_{s=1}^S \sum_{f=1}^{T-1} \sum_k \zeta_{k,f}^{(s)}$.

Priority 0 (lowest) — Discounted realized tax cost. For each filtration $f \in \{1, \dots, T\}$, realized gain/loss is

$$\text{tc}_f^{(s)} = \tau \sum_{(i,j) \in \mathcal{E}_f} \sigma_{i,j,f}^{(s)} (p_{k(i,j),f}^{(s)} - c_{i,j}^{(s)}). \quad (7)$$

Losses contribute negatively, incentivizing tax-loss harvesting. We apply a geometric discount γ^{f-1} that down-weights tax cost incurred later in the horizon, analogous to the discount factor in reinforcement learning:

$$J_0 = \frac{1}{S} \sum_{s=1}^S \sum_{f=1}^T \gamma^{f-1} \text{tc}_f^{(s)}. \quad (8)$$

With $\gamma = 1$ the objective is the undiscounted expected tax bill in dollars. Setting $\gamma < 1$ is intended to incentivize earlier loss realization, since an early-period harvested loss is weighted more heavily than a deferred one; Section 4 examines the closed-loop consequences. A 5% relative MIP gap and a five-minute time limit are applied to this stage via Gurobi’s per-objective sub-environment; the deviation stages keep Gurobi’s default tight gap.

3 Robust Model Predictive Controller

Block bootstrap. Let $\{r_t\}_{t=1}^H$ with $r_t \in \mathbb{R}^N$ be the observed monthly return-vector series over a historical window. Form all overlapping blocks of length B :

$$\mathcal{B} = \{(r_t, r_{t+1}, \dots, r_{t+B-1}) : t = 1, \dots, H - B + 1\}. \quad (9)$$

For each scenario $s = 1, \dots, S$, draw $\lceil (T - 1)/B \rceil$ blocks i.i.d. uniformly with replacement, concatenate, and trim to exactly $T - 1$ return vectors $(r_2^{(s)}, \dots, r_T^{(s)})$, indexed by the filtration they advance to. Stitch onto the observed first-filtration price $p_1 \in \mathbb{R}^N$:

$$p_1^{(s)} = p_1, \quad p_f^{(s)} = p_{f-1}^{(s)} \odot (1 + r_f^{(s)}), \quad f = 2, \dots, T. \quad (10)$$

The block structure preserves short-range cross-asset dependence (factor co-movement, volatility clustering) that an i.i.d. resample would destroy, while remaining nonparametric [1]. Scenarios share p_1 because it is observed at the controller’s invocation; the scenario fan opens at $f = 2$.

Information-pattern (non-anticipativity) constraint. The bootstrap produces S distinct price paths but the controller must commit to a single first-stage action before the scenarios diverge. With $f = 1$ the first decision filtration, enforcing

$$x_{i,j,1}^{(s)} = x_{i,j,1}^{(s')} \quad \forall s, s' \in \{1, \dots, S\}, \quad (i, j) \in \mathcal{L}_1, \quad (11)$$

pins the post-decision lot vector at $f = 1$ equal across all scenarios. Combined with the lot dynamics, this forces the first-stage sells from starting lots $\sigma_{i,0,1}^{(s)}$ and first-stage buys $\beta_{k,1}^h(s)$ (which initialize column $j = 1$) to be identical across s . From $f \geq 2$ onward, decisions are scenario-specific—the full-recourse stage. The construction of the `RMPController` reproduces the standard two-stage stochastic program [3].

Receding-horizon backtester. The `Backtester` implements an execute-and-re-plan scheme. Let $\{\hat{p}_t\}_{t=0}^{T_{\text{act}}-1}$ be the realized monthly price series along which the closed-loop policy is evaluated. At each step $t = 0, \dots, T_{\text{act}} - 2$: (i) construct a fresh `RMPController` with current positions and redraw the bootstrap from the rolling historical window; (ii) solve the multi-scenario program once—the information-pattern constraint guarantees the first-stage action plan ($\sigma_{i,0,1}$ and $\beta_{k,1}^h$) is identical across scenarios; (iii) execute *only* those first-stage trades against realized prices $\hat{p}_t$, updating lots and opening new ones with cost basis $\hat{p}_{k,t}$; (iv) advance $t \rightarrow t + 1$. All scenario-dependent decisions at $f \geq 2$ are discarded—they were only for planning. Re-solving from the updated state at $t + 1$ replaces them with actions that incorporate the realized price and lot structure, recovering the standard MPC closed-loop policy applied to a stochastic, mixed-integer plant.

4 Results

Experiment setup. We transition \$1,000,000 in AAPL (cost basis \$500,000) into an equal-weight portfolio of the top-20 S&P 500 stocks (5% per ticker), rebalanced monthly over January–December 2024 with a flat 30% tax rate. An immediate full sell to target would realize $(1,000,000 - 500,000) \times 0.95 = \$475,000$ in capital gains and \$142,500 in tax—the myopic baseline. To isolate each layer’s contribution we run three experiments: the *prescient* case (exact future prices known), the *MPC* $t = 0$ solve (one-shot planning under simulated scenarios), and a *monthly backtest* (receding-horizon execution). Progress is tracked via transition percentage $\text{tp}_f \in [0, 1]$, with 1.0 a perfect weight match. For ticker k at filtration f , let $o_{k,f} = \max(0, w_{k,f} - w_k^*)$ and $u_{k,f} = \max(0, w_k^* - w_{k,f})$ denote per-ticker overweight and underweight; then

$$\text{tp}_f = 1 - \max\left(\sum_k o_{k,f}, \sum_k u_{k,f}\right). \quad (12)$$

Results are shown in Figure 2, run with intermediate band $\delta = 5\%$ and discount factor $\gamma = 1$ unless stated otherwise.

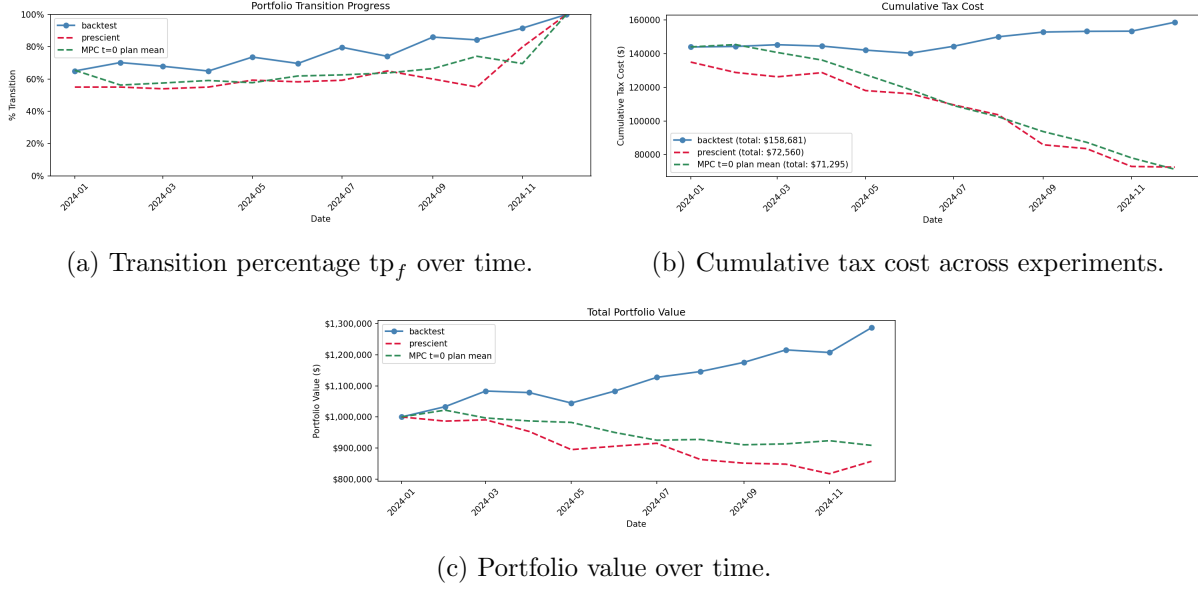


Figure 2: Results for the AAPL-to-equal-weight-S&P 500 transition, January–December 2024.

Prescient case. We solve `StoxOptimizer` with realized 2024 prices as a single scenario, establishing a lower bound on achievable tax cost under full information. This exhibits look-ahead bias: the optimizer exploits knowledge of future price paths to time loss harvesting optimally. As shown by the red lines, it first sells AAPL to satisfy the δ band, reaching $tp \approx 60\%$, then harvests losses in subsequent periods, ending with a cumulative tax cost of \$72,560 and full transition by December.

MPC $t = 0$ planning. We run a single `RMPCController` solve using block-bootstrapped scenarios (block length $B = 3$ months, historical window January–December 2023, $S = 10$ scenarios). The green line, representing the mean across scenarios. This is the plan according to `RMPCController` at $t=0$.

Monthly backtest. We run `Backtester` with monthly rebalancing throughout 2024, maintaining a rolling one-year historical window for each bootstrap draw ($S = 10$). The blue lines show good transition progress but negligible tax-loss harvesting activity, yielding a cumulative tax cost (\$158,681) that exceeds even the myopic immediate-sell baseline. The gap between the prescient lower bound and the realized backtest is the cost of the prescient optimizer’s look-ahead bias: 2024 was a rising market, and a bootstrap calibrated on 2023 returns generates scenarios that misrepresent the realized path, misleading the controller into trades that do not pay off out of sample.

Tuning the discount factor γ and band δ . We implemented the discounted factor γ (Eq. 8) and swept it jointly with the intermediate band δ ; Table 1 reports realized backtest tax cost. Two effects emerge. First, widening δ grants the optimizer more freedom to drift from target immediately, which it spends on additional turnover, chasing unrealized tax loss harvesting opportunities; the extra trading realizes more gains and raises tax cost (\$158,681 at $\delta = 0.05$ versus \$184,905

at $\delta = 0.2$, both at $\gamma = 1$). Second, while $\gamma < 1$ does produce early harvesting, it also leads the optimizer to defer gain realization to the cheap tail of the horizon, because gains are positive contributions to J_0 , discounting later periods makes total discounted tax cost smaller—procrastinating until it is forced to dump the remaining appreciated lots at the terminal period to satisfy the terminal-deviation objective. This pushes tax cost higher still (\$233,153 at $\delta = 0.2, \gamma = 0.5$ and \$227,128 at $\delta = 1, \gamma = 0.5$).

Configuration	Realized tax	Note
Immediate full sell	\$142,500	baseline
$\delta = 0.05, \gamma = 1$	\$158,681	higher tax cost than baseline
$\delta = 0.2, \gamma = 1$	\$184,905	higher turnover $\rightarrow$ higher tax cost
$\delta = 0.2, \gamma = 0.5$	\$233,153	defer gain $\rightarrow$ partial dump at last period
$\delta = 1, \gamma = 0.5$	\$227,128	procrastination $\rightarrow$ terminal dump

Table 1: Realized backtest tax cost under different intermediate band δ and discount factor γ

5 Conclusions and Future Work

We attribute the portfolio’s poor tax-loss harvesting to two causes. First, the portfolio setup itself offers little to harvest: the position carries a large embedded gain in AAPL, which rose steadily through 2024, so depreciated lots rarely appear, and there is simply nothing to realize. This is a property of the instance, not the framework, and warrants testing across a range of starting portfolios—higher lot diversity, mixed embedded gains and losses, different market regimes—to assess where the framework adds value. Second, forecast quality dominates harvesting effectiveness, and a robust controller cannot fully compensate for it. Accurate price prediction lets the optimizer time gain deferral and loss realization; under misleading forecasts, the benefit of deferring gains evaporates, and a myopic immediate loss harvesting policy is hard to beat. The prescient lower bound (\$72,560) confirms the framework works when the forecast is right—the gap to the backtest is the price of getting the forecast wrong.

Two directions follow. First, **forecast quality**: pair the robust controller with an alpha model, so the scenario distribution reflects a genuine view on returns rather than an unconditional resample of a recent window. Second, **scaling scenarios**: richer scenario sets are needed for robust planning to be meaningful, but the two-stage program has a mixed-integer master *and* mixed-integer recourse, so naive scaling is intractable. Specialized branch-and-Benders-cut decomposition [4, 2] is the natural route to larger S .

6 References

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